Library Management System

October 1, 2023

1 Introduction

This document describes the implementation of a Library Management System. The system supports different user roles (Student, Faculty, Librarian) and allows users to borrow and return books, manage fines, and handle library data persistence.

2 Classes

- **User**: Abstract base class for different user roles.
- **Student**: Derived class representing a student user.
- **Faculty**: Derived class representing a faculty user.
- **Librarian**: Derived class representing a librarian user.
- **Book**: Class representing a book in the library.
- **File_Handler**: Class for handling file operations related to books and users.
- **Account**: Class representing a user's account, managing borrowed books and fines.
- **Authentication**: Class for handling user authentication and credentials.
- **Library**: Class representing the library, managing books, users, and accounts.
- **LibrarySystem**: Class for managing the overall library system, including user interactions.

3 Functions

- **main**: Entry point of the program, initializes and starts the library system.

4 Usage

1. Compile and run the program.
2. Follow the menu prompts to interact with the library system.

5 File Structure

- `borrowed_books.csv`: Stores information about borrowed books.
- `books.csv`: Stores information about books in the library.
- `users.csv`: Stores information about users in the library.
- `credentials.csv`: Stores user credentials for authentication.

6 Dependencies

- `<iostream>`
- `<vector>`
- `<string>`
- `<chrono>`
- `<algorithm>`
- `<map>`
- `<fstream>`
- `<sstream>`

7 How to Operate

1. Compile the program:
   ```
   g++ -o library_system ASSIGNMENT.cpp
   ```
2. Run the executable:
   ```
   ./library_system
   ```
3. Follow the menu prompts to interact with the library system:
   - Login with your user ID and password.
   - Depending on your role (Student, Faculty, Librarian), you will see different menu options.

- Students and Faculty can view available books, borrow books, return books, view borrowed books, and pay fines.
- Librarians can add books, remove books, display all books, display all users, and add new users.
- Use the menu options to perform the desired actions.
- Logout or exit the system when done.

7.1 Default Passwords

- Students: `pass123`
- Faculty: `prof123`
- Librarians: `lib123`

Change your password after the first login for security reasons.

8 Example Usage

1. Start the program and login as a librarian:
   - User ID: `lib001`
   - Password: `lib123`
2. Add a new book to the library:
   - Title: `C++ Programming`
   - Author: `Bjarne Stroustrup`
   - Publisher: `Addison-Wesley`
   - Year: `2013`
   - ISBN: `9780321563842`
3. Add a new student user:
   - Name: `John Doe`
   - User ID: `stu001`
   - Role: `Student`
4. Logout and login as the new student user:
   - User ID: `stu001`
   - Password: `pass123`
5. Borrow the book `C++ Programming`:
   - Enter ISBN: `9780321563842`

6. View borrowed books and pay any fines if applicable.
7. Return the borrowed book:
    - Enter ISBN.